

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. Examination, 2024
MATH – 1105, Differential and Integral Calculus

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Evaluate the following limits:

4

$$i) \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} \quad ii) \lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} - \cos \theta - \sin \theta}{(4\theta - \pi)^2}$$

- b) Find the maximum and minimum values of the function $f(x, y) = 3x + 4$ on the circle $x^2 + y^2 = 1$

5

c) If $f(x) = \begin{cases} x & \text{when } x \leq 0 \\ x & \text{when } 0 < x < 1 \\ 1-x & \text{when } x \geq 1 \end{cases}$

3+2

Show that the function $f(x)$ is continuous at $x = 0$ and discontinuous at $x = 1$.

2. a) Differentiate $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ with respect to $\tan^{-1} x$.

4

- b) Prove that ,

$$\int_0^{\frac{\pi}{2}} \sin^p \theta \cos^q \theta d\theta = \frac{\Gamma(\frac{p+1}{2}) \Gamma(\frac{q+1}{2})}{2\Gamma(\frac{p+q+2}{2})}$$

6

- c) If $y = x^{2n}$ then show that $y_n = 2^n \{1.3.5 \dots (2n-1)\} x^n$ where $n \in N$

4

3. a) State and prove Leibnitz's theorem.

1+4

- b) If $y = e^{a \sin^{-1} x}$ then show that $(1-x^2)y_2 - xy_1 - a^2 y = 0$

5

- c) Find the radius of the curvature of the asteroid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ at any point.

4

4. a) Integrate :

3

$$\int \frac{1}{\sqrt{1-x^2} \sqrt{\sin^{-1} x}} dx$$

- b) Solve the following by using L' hospitals rule:

6

$$i) \lim_{x \rightarrow 0} \left[\frac{e^x - e^{\sin x}}{x - \sin x} \right] \quad ii) \lim_{x \rightarrow 0} (\sin x)^x \quad iii) \lim_{x \rightarrow 0} x^2 \ln(x)$$

- c) Show that $\int_0^{\frac{\pi}{2}} \sin^6 \theta \cos^3 \theta d\theta = \frac{2}{63}$

3

- d) Find out the Maclaurin's series of $1/(1+x)$

2

5. a) What is Taylor's infinite series? Using Taylor's theorem, prove that

2+3

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5(1+c)^5}, \text{ where } 0 < c < x$$

- b) Show that

5

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

- c) Locate the relative maxima and minima of $f(x) = \frac{4}{x} + \frac{36}{2-x}$; Also find the extremum value.

4

6. a) The volume of spherical balloon is increasing at the rate of $12 \text{ cm}^3 \text{ sec}^{-1}$. Find the rate of change of its surface at the instant when its radius is 6 cm. 4
- b) By (δ, ϵ) definition, prove that $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6$ 2
- c) Find the area of the solid formed by the revolution of the cardioid of the curve $r = a(1 + \cos \theta)$ about the initial line. 4
- d) Evaluate: i) $\int \frac{(2x^2 - 1)}{(x+1)^2(x-2)} dx$. ii) $\int_0^{\frac{\pi}{2}} \cos^3 x (\sin x)^{\frac{1}{4}} dx$ 4

7. a) Evaluate, 2

$$\lim_{n \rightarrow \infty} \left[\frac{n}{n^2 + 1^2} + \frac{n}{n^2 + 2^2} + \dots + \frac{n}{n^2 + n^2} \right]$$

- b) Verify the mean value theorem for the function $f(x) = \frac{1}{x}$ on $(-1, 1)$ 2
- c) Find the equation of the tangent and normal at the point (x, y) of the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$. 4
- d) Evaluate (any three): i) $\int_{\alpha}^{\beta} \frac{1}{\sqrt{(x-\alpha)(\beta-x)}} dx$ ii) $\int_0^1 \frac{1}{(1+x^2)\sqrt{(1-x^2)}} dx$ iii) $\int_0^1 \frac{\sin^{-1} x}{\sqrt{(1-x^2)}} dx$ 6
- iv) $\int_0^{\frac{\pi}{2}} \frac{1}{3+2\sin x} dx$